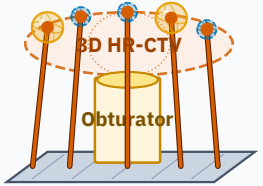
a Clinical bottleneck in MRI-guided GYN HDR brachytherapy vs. NeedleFinder 2026 workflow



3D template (15–30 6F tubes)

Manual OR tracing (status quo)

- 45–50 min slice-by-slice clicking
- Signal voids on 3T T2W (1.6–5 mm)
- Tip discrepancy: 3.24 mm median

NeedleFinder 2026 (3D Slicer)

- **4.15 s CPU (2.67 s GPU): 89.4% F1**
- Shaft 0.47 mm · Tip 2.24–2.27 mm
- **+18 s* 2-tip check → 2.8% |ΔD90|**

b Three physical clues separate plastic catheters from pelvic mimics across scales

Clue 1: 3D Dark-Core Symmetry

6 diameters × 4 radii (1.35–3.3 mm)



[+] Catheter: bright rim both sides

[-] Organ edge: 1-sided step (min=0)

Clue 2: Centerline Peaks

Touching 6F pair (2.2 mm apart)



[-] Binary mask: merges into 1 blob

[+] Peak head: 2 sharp local maxima

Clue 3: 3D Bundle & 1D Tip

Whole-bundle graph + 91-step tip



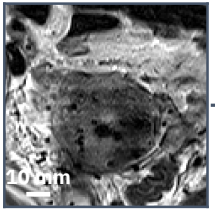
[+] Smooth 3D shaft + tip stop

[-] Pelvic vessel: bends / outside gate

c End-to-end visual progression on dense clinical implant (Case 017: 26 catheters + central obturator, 4.15 s CPU)

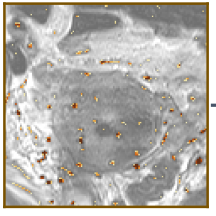
1. Input 3T T2W MRI

Native anisotropic grid



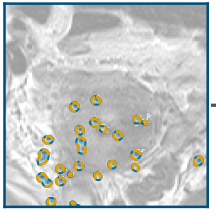
2. Stage 0 Symmetry

Dark-core response map



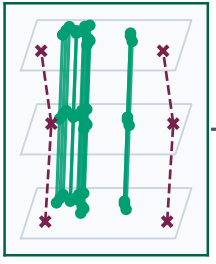
3. Stage 1 3D ResUNet

Peaks + tilt + obturator



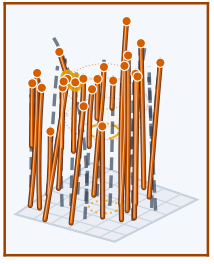
4. Stage 2A/B Graph

Triplet MPNN + stitch



5. Stage 2C 3D Output

23 pred. / 26 ref. + tips



3D bundle key: -- Expert reference trajectory (charcoal dashed) — NeedleFinder 2026 shaft (vermilion) [o] Predicted 3D distal tip